

PROJECT DESIGN PHASE-II: TECHNOLOGY STACK

Date	October 03, 2026	Maximum Marks	4 Marks
Team ID	SWTID-2026-6919	Project Name	Script-Controlled ACL - Restrict Record Access Based on Field Value

TECHNICAL ARCHITECTURE & GUIDELINES

Architectural Guidelines & Deliverable Requirements:

- **Application Logic & Process Flow:** Include all server-side and client-side logic blocks controlling ACL execution.
- **Infrastructural Demarcation:** Clearly differentiate local development components vs. ServiceNow Cloud Platform infrastructure.
- **External Interfaces:** Define Integration endpoints, REST/SOAP APIs, and identity providers.
- **Data Storage:** Map core relational tables, field-level access definitions, and attachment storage.
- **Security & Access Control:** Specify role-based (RBAC) and script-based security constraints.

TABLE 1: TECHNICAL ARCHITECTURE & COMPONENT STACK

S.No	Component	Description	Technology Stack
1	User Interface	How user interacts with application e.g., Web UI, Service Portal, Workspace, or Mobile App.	HTML5, CSS3, JavaScript / AngularJS, ServiceNow Next Experience UI
2	Application Logic - 1	Core business process logic for record handling and field-level state validation.	JavaScript (ES6), ServiceNow Server-Side Business Rules
3	Application Logic - 2	Script-controlled Access Control List (ACL) evaluation rules based on dynamic field values.	ServiceNow Security Engine, Server-Side JavaScript (ACL Scripts)
4	Application Logic - 3	Reusable script utilities for role validation and field permissions checking.	ServiceNow Script Includes (GlideRecord, GlideUser)
5	Database	Relational schema storing system tables, user records, roles, and field values.	ServiceNow Relational Database / MariaDB Engine
6	Cloud Database	High-availability managed multi-instance cloud database infrastructure.	ServiceNow Enterprise Cloud Infrastructure
7	File Storage	Storage mechanism for system logs, file attachments, and audit history.	ServiceNow Attachment Table (sys_attachment) / Cloud Storage
8	External API - 1		

S.No	Component	Description	Technology Stack
		REST API interface for third-party integration and automated record query/update.	ServiceNow Table API / REST API Explorer
9	External API - 2	Single Sign-On (SSO) and Identity Provider integration for user authentication.	OAuth 2.0 / SAML 2.0 / Azure AD Integration
10	Machine Learning Model	Predictive intelligence for automated ticket categorization and risk evaluation (if applicable).	ServiceNow Predictive Intelligence / Machine Learning Engine
11	Infrastructure (Server / Cloud)	Application deployment model across development, test, and production environments.	ServiceNow SaaS Cloud Infrastructure (Multi-Tenant Architecture)

TABLE 2: APPLICATION CHARACTERISTICS

S.No	Characteristics	Description	Technology Stack
1	Open-Source Frameworks	Web standards and open frameworks utilized within UI pages, Service Portal widgets, and client scripts.	AngularJS, Bootstrap 3/4, JavaScript (ES6+), HTML5/CSS3
2	Security Implementations	Multi-layered security including Contextual Security (ACLs), Role-Based Access Control (RBAC), Scripted Evaluation, and Elevated Roles (security_admin).	SHA-256, TLS 1.3 Encryption, ServiceNow ACL Engine, OWASP Guidelines
3	Scalable Architecture	Multi-instance SaaS architecture supporting horizontal scaling, instance clustering, and isolated database instances.	ServiceNow Advanced High Availability (AHA) Architecture
4	Availability	High availability guarantees with active-active redundant enterprise data centers, automatic failover, and load balancing.	ServiceNow HA Cloud, F5 Load Balancers, Distributed Data Centers
5	Performance	Design considerations for high throughput: in-memory caching, indexed query optimization, asynchronous processing, and Content Delivery Networks.	ServiceNow In-Memory Cache, Indexed Queries, CDN Integration

REFERENCES

- C4 Model for Software Architecture: <https://c4model.com/>
- ServiceNow Developer & Access Control Documentation: <https://developer.servicenow.com/>
- IBM Cloud Architecture Center: <https://www.ibm.com/cloud/architecture>
- AWS Architecture Center: <https://aws.amazon.com/architecture>
- Technical Architecture Diagram Guidelines: <https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>